Sheet: bom

Component	Group	g/cell	kg/kWh	Mass %	Basis / note
NMC811 active material	Cathode	18.0163	0.71	35.67	tw 96/2/2 cathode composition split = LITERATURE-STANDARD ESTIMATE (parameter set gives lumped density 3262 kg/m3 only)
Carbon black conductive additive	Cathode	0.3753	0.0148	0.74	est. 2 wt%
PVDF binder	Cathode	0.3753	0.0148	0.74	est. 2 wt%
Graphite active material	Anode	10.9476	0.4314	21.67	tw 96/2/2 anode composition split = LITERATURE-STANDARD ESTIMATE (lumped 1657 kg/m3)
Conductive additive	Anode	0.2281	0.009	0.45	est. 2 wt%
Binder (CMC/SBR class)	Anode	0.2281	0.009	0.45	est. 2 wt%
Al current collector (8 um)	Current collectors	2.2183	0.0874	4.39	2700 kg/m3 std
Cu current collector (6 um)	Current collectors	5.5212	0.2176	10.93	8960 kg/m3 std
Separator (PP/PE class, 8 um dry)	Separator	0.1729	0.0068	0.34	397 kg/m3 from Chen2020 set
Electrolyte fill	Electrolyte	12.4286	0.4898	24.61	pore volume 10.36 cm3 x 1.2 g/cm3 = LITERATURE ESTIMATE (parameter set lacks electrolyte density - contract caliber excludes it)
CONTRACT-CALIBER TOTAL (stack, excl electrolyte)		38083	1.5008	100	= 666.3 Wh/kg evaluated value
BOM-CALIBER TOTAL (incl electrolyte fill)		50.512	0.002	100	~502 Wh/kg packaging reality incl. electrolyte only (still no can/tabs)
Electrolyte transport spec (parameter bridge)					sigma 1.5 S/m (est), t+ 0.6 (est), D 5.0e-10 m2/s (est); NOT DFT/MD-endorsed
Cell format					assembled stack 242.12 um x 0.1027 m2; can/tabs not modeled